

The Function Q_X Is Not Local with Respect to Connected Compact Neighbourhoods

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Status: full counterexample to Open Question 8. There are a connected compact plane set X , a point $z_0 \in X$, and a connected compact neighbourhood N of z_0 in X such that

$$Q_N(z_0) = \infty \quad \text{but} \quad Q_X(z_0) = 1.$$

Indeed, X may be the closed unit disk. Thus the implication asked for by Dales and Feinstein is false, even for a convex ambient set.

1 Source question

For a connected compact plane set Y and $z \in Y$, Dales and Feinstein [1] define

$$Q_Y(z) = \sup \left\{ \frac{|f(z) - f(y)|}{|z - y|} : y \in Y \setminus \{z\}, f \in D^{(1)}(Y), |f'|_Y \leq 1 \right\}.$$

Their Open Question 8 on arXiv PDF page 31 asks whether $Q_N(z_0) = \infty$ for a connected compact neighbourhood N of z_0 in X must imply $Q_X(z_0) = \infty$.

- (8) (Local behaviour of the function Q_X .) Let X be a connected, compact plane set, and let $z_0 \in X$. Suppose that z_0 has a connected, compact neighbourhood N in X such that $Q_N(z_0) = \infty$. Is it necessarily true that $Q_X(z_0) = \infty$?

Figure 1: Open Question 8 on source PDF page 31.

2 Counterexample

Theorem 1. *Let $X = \overline{\mathbb{D}}$, let $z_0 = 0$, and let $E = \{z \in \mathbb{C} : |z| \leq 1/4\}$. There is a non-rectifiable Jordan arc $J \subset X$ with endpoints $q = 1/4$ and $w = 3/4$ such that $J \cap E = \{q\}$. For*

$$N = E \cup J$$

we have

$$Q_X(0) = 1, \quad Q_N(0) = \infty.$$

In particular, the answer to Open Question 8 is no.

Such an arc is elementary to arrange: leave q along the real segment to a point outside E , then continue along a sufficiently small scaled Koch arc ending at w . The construction can be kept in a thin rectangle contained in the open unit disk and disjoint from E except at q . Adding the initial segment preserves the Jordan property and leaves the total length infinite.

3 Idea of the proof

The ambient disk has every straight segment available, so bounded derivative forces the sharp Euclidean Lipschitz estimate and $Q_X(0) = 1$. The thin neighbourhood N , however, contains a non-rectifiable arc. A lemma already proved in the source constructs functions on an arbitrarily long Jordan arc whose derivative is uniformly bounded but whose endpoint oscillation is as large as desired, with derivative zero at both endpoints. The zero endpoint derivative lets us patch each such function constantly across the small disk E . These functions belong to $D^{(1)}(N)$ but cannot arise by restriction from the full disk. This mismatch makes $Q_N(0)$ infinite while $Q_X(0)$ stays minimal.

4 Proof

We use the following source result.

Lemma 1 (Dales–Feinstein, Lemma 10.4). *Let $a, b \in \mathbb{C}$, let $A > 0$, and let γ be a Jordan path from a to b whose length is greater than A (infinite length is allowed). Then there is $h \in D^{(1)}(\gamma^*)$ such that*

$$0 < |h'|_{\gamma^*} \leq 3, \quad h'(a) = h'(b) = 0, \quad |h(a) - h(b)| > A/2.$$

First compute $Q_X(0)$. Let $f \in D^{(1)}(X)$ with $|f'|_X \leq 1$ and let $u \in X$. The straight segment $t \mapsto tu$ lies in X . The one-variable fundamental theorem of calculus, applied to $t \mapsto f(tu)$, gives

$$|f(u) - f(0)| \leq \int_0^1 |uf'(tu)| dt \leq |u|.$$

Thus $Q_X(0) \leq 1$. The coordinate function $f(z) = z$ gives equality for every $u \neq 0$, hence

$$Q_X(0) = 1. \tag{1}$$

We now prove that $Q_N(0) = \infty$. Fix $A > 0$. Since J has infinite length, Lemma 1 gives $h_A \in D^{(1)}(J)$ such that

$$|h'_A|_J \leq 3, \quad h'_A(q) = 0, \quad |h_A(q) - h_A(w)| > A/2.$$

Define $F_A : N \rightarrow \mathbb{C}$ by

$$F_A(z) = \begin{cases} h_A(q), & z \in E, \\ h_A(z), & z \in J. \end{cases}$$

This is well defined because $E \cap J = \{q\}$. It belongs to $D^{(1)}(N)$. Away from q this follows separately on the two compact pieces. At q , the difference quotient is identically zero along E and converges to $h'_A(q) = 0$ along J ; hence $F'_A(q) = 0$. The same two-piece observation, together with continuity of h'_A and $h'_A(q) = 0$, shows that F'_A is continuous on N . Therefore

$$|F'_A|_N \leq 3.$$

Set $G_A = F_A/3$. This is an admissible test function in the definition of $Q_N(0)$, and $G_A(0) = h_A(q)/3$. Consequently

$$Q_N(0) \geq \frac{|G_A(0) - G_A(w)|}{|w|} > \frac{A}{6|w|}.$$

Since A is arbitrary and $w = 3/4$ is fixed,

$$Q_N(0) = \infty. \tag{2}$$

The set N is compact and connected. It is a neighbourhood of 0 in X because it contains the relative open set $X \cap \{z : |z| < 1/4\}$. Equations (1)–(2) therefore give exactly the counterexample asserted in Theorem 1.

5 Strength and scope

The example uses no pathology in the ambient geometry: X is convex, pointwise regular, polynomially convex, and has nonempty interior. The value $Q_X(0) = 1$ is the smallest possible value. The construction is also robust: any convex compact set containing a small convex core and an attached non-rectifiable Jordan arc gives the same phenomenon after rescaling.

The word “neighbourhood” is used in its standard topological sense: N contains a relatively open neighbourhood of z_0 in X . No claim is made for stronger variants in which N is required, for example, to be the closure of a relatively open connected set.

The run’s registry, solution, attempt, and proof-gap indexes were searched by arXiv id, title, “local behaviour Q_X ,” “ $Q_N(z_0)$,” “compact neighbourhood,” and “non-rectifiable Jordan arc.” Exact-phrase and topic searches found no later resolution of Question 8. The only imported ingredient is source Lemma 10.4, stated in full above; the attachment and both Q computations are new deductions.

References

- [1] H. G. Dales and J. F. Feinstein, *Normed algebras of differentiable functions on compact plane sets*, Indian J. Pure Appl. Math. **41** (2010), 153–187; arXiv:1501.03986.